MOST APPS TELL YOU WHAT YOU DID. OURS TELLS YOU WHEN TO STOP.

An AI coach that watches you train with **your phone camera** and **your smartwatch**. It recognizes the exercise, checks your form, compares today with **your own past sessions**, and tells you to stop before fatigue turns into injury.

1 HOW IT WORKS

CAPTURE
Phone + watch
Phone films you; each frame becomes a skeleton of 18 joints (MediaPipe, not trained by us). Watch sends heart rate + wrist motion.

→

MODEL 1
What & how many
Which exercise, and where each rep starts and ends.

→

MODEL 2
Good or bad
Judges each rep from Model 1: good, or which mistake (knees inward, not low enough...).

→

MODEL 3 · OUR CONTRIBUTION
When to stop
Rep speed + form + heart rate, compared with **your** past sessions.

→

VOICE
The verdict
"Keep going", "Slow down", or "Stop".

> **Big data side:** Kafka streams the live data to the laptop (≈ 300 messages per second). Spark processes every stored session to build each person's history.

2 THREE DATASETS, THREE JOBS

RECOGNIZE · COUNT · HEART RATE

MM-FIT

70.8 M sensor rows · **6,160** reps · **10** people

21 sessions, 10 exercises. Camera + watches + phone recorded at the same time. 3.8 GB.

Missing: no good/bad label on reps.

GOOD REP OR BAD REP

EC3D

371 labelled reps · **29,789** 3D frames

8 mistakes

Squat, lunge, plank in 3D (25 joints), 4 people. Full file: 62,387 frames × 4 cameras.

Missing: heart rate. Only 4 people.

FATIGUE OVER WEEKS

OUR RECORDINGS

2 people · **3–6** weeks · **1–10** effort score

The two of us, Poco X3 Pro + Galaxy Watch 7, effort score after each set.

The only data that follows the same person over time.

3 THE 4 RESEARCH QUESTIONS, IN PLAIN WORDS

RQ1 Does it recognize the exercise and count reps on a person it has never seen?

A coach that only works on its training people is useless.

Answer looks like: "92 % correct on new people, 0.3 reps off per set."

RQ2 Does the watch's heart rate detect fatigue better than movement alone?

The teacher asked us to use the watch's vital data. We train Model 3 with and without heart rate.

"With heart rate, the warning comes 2 reps earlier." Or "it doesn't help". Both are results.

RQ3 Is comparing you with your own past sessions better than comparing you with an average person? MAIN IDEA

People are very different; one threshold can't fit everyone.

Your squats are usually 1.4 s, today 1.8 s → warning, even if 1.8 s is normal for someone else.

RQ4 Does it still work on our own phone and Galaxy Watch 7, outside the dataset?

The "useful in real life" test.

"90 % on the dataset, 80 % on our devices: we lose 10 % because of camera quality and angle."

4 WHAT WE ALREADY MEASURED

> Quick tests on the real datasets. Present them as **first measurements**, not final results.

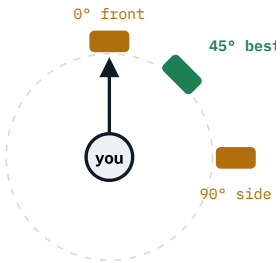
45 / 64
squat sets end with **slower reps**: fatigue is visible
Lunges: no clear effect, so speed alone isn't enough → we add heart rate.

63 / 64
squat sets where reps were **counted exactly** from the skeleton
62 / 62 for lunges. Our rep timings are reliable.

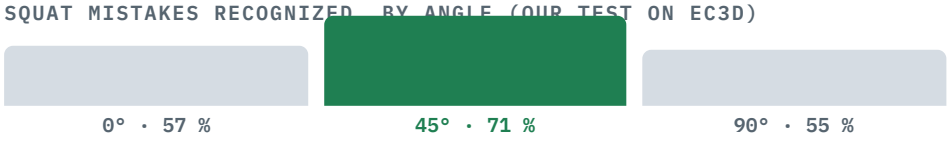
53–90 %
mistakes recognized on a **new person** (squat → plank)
Chance is 20–33 %. EC3D authors reach 90.9 %: our target.

123–188
peak heart rate (bpm) **between people**
One fixed threshold can't work → personal baseline (RQ3).

5 WHERE TO PUT THE PHONE



1. **About 45° in front** of the user. From the front you can't judge depth; from the side one leg is hidden.
2. Phone at **hip height**, **2–3 m** away, whole body in the frame.
3. **Same spot every session**, so a change in the numbers means fatigue, not a moved camera.



6 WHEN THE TEACHER SAYS..., WE ANSWER

✕ *"Comparing live frames to the dataset is not training."*	✓ We train 3 models ourselves: recognition, form check, fatigue and risk.
✕ *"Camera + watch beats watch alone, that's obvious."*	✓ Agreed. It's only a sanity check, not our claim. Our claims are RQ2 (heart rate) and RQ3 (personal history).
✕ *"Posture detection already exists."*	✓ Yes, so we reuse MediaPipe for it and spend our training on fatigue and risk.
✕ *"Who guarantees the reps in the dataset are correct?"*	✓ EC3D: every mistake was performed on instruction and each sequence was checked by hand. MM-Fit is never used to judge form.
✕ *"You have no injury labels."*	✓ Right, and we never claim to predict injuries. We detect fatigue and form breakdown, the warning signs, and recommend stopping.
✕ *"EC3D has only 4 people."*	✓ We test the strictest way (train on 3, test on the 4th), augment with 3D views from any angle, and add our own reps as a test set.
✕ *"3.8 GB is not big data."*	✓ 70.8 M rows already, and the hard part is speed and variety: 4 live streams at 1–100 Hz to sync, history growing every session.
✕ *"Two people is not a dataset."*	✓ It's a proof of concept: each of us is compared with our own history. No public dataset follows one person for weeks.

TO SETTLE TOGETHER TONIGHT

How do we get **live heart rate from the Galaxy Watch 7** without writing a Kotlin app? We need one clear answer: which free app or export method.

7 THE PRESENTATION APP

Open **presentation/index.html** in Chrome. 9 slides, about 8½ minutes. Extra controls are hidden on purpose:

- ← → — Previous / next slide
- **B** — Appendix with all details and the "Hard questions" slide. Or double-click the logo.
- **T** — Show and start a timer. Press again to hide.
- **F** — Fullscreen

A REP COUNTER THAT BECOMES A COACH THAT PROTECTS YOU.

3 models we train ourselves
2 devices people already own
1 personal history